SE375 SYSTEM PROGRAMMING SPRING 2025-2026
Laboratory Assignment 0
Java Warm-up

February 20, 2026

Discrepancies in the Resident List

The municipality had ordered a complete revision of the people living in each apartment. Knowing the list of residents is important and counting them prone to errors. Some residents may have died, and others may have moved and so on.

To be as accurate as possible, the municipality assigned this task to three different officials. Each of them was to independently record the list of residents and send it to the municipality. Each resident has an ID, and the list would list the ID numbers. The officials were expected to arrange the ID numbers in ascending order in their lists. ID numbers are unique and not duplicated in any list.

On receiving the lists, the municipality realized that there were discrepancies - the three lists were not identical. The municipality decided to go with the majority. That is, that decided to construct the final list including only those ID numbers that appeared in at least 2 out of the 3 lists. For example, if the three lists were

```
28  30  42  63  91
21  28  35  63  91  92
21  28  30  63  91
```

then the final list compiled by the municipality would be:

```
21  28  30  63  91
```

The ID numbers 35, 42 and 92 which appeared in only one list each do not figure in the final list. Your task is to write a program that produces the final list from the three given lists.

Input format:

Content of the file	Explanation
`5 6 5`	Number IDs in each list
`28 30 42 63 91`	Space seperated IDs in List 1 in acending order
`21 28 35 63 91 92`	Space seperated IDs in List 2 in acending order
`21 28 30 63 91`	Space seperated IDs in List 3 in acending order

Output:

```
5                          /* the number of voters in the final list. */
21 28 30 63 91             /* the list of IDs in the final list, in ascending order. */
```

IMPLEMENTATION NOTES:

1. Read the data from a file
2. Use HashMap and ArrayList from the Java Collection framework in solving this problem. No other data structure will be accepted.

BONUS:

If time permits, assume that the input data is not sorted. First sort it using *sort* from the *Collections* class.